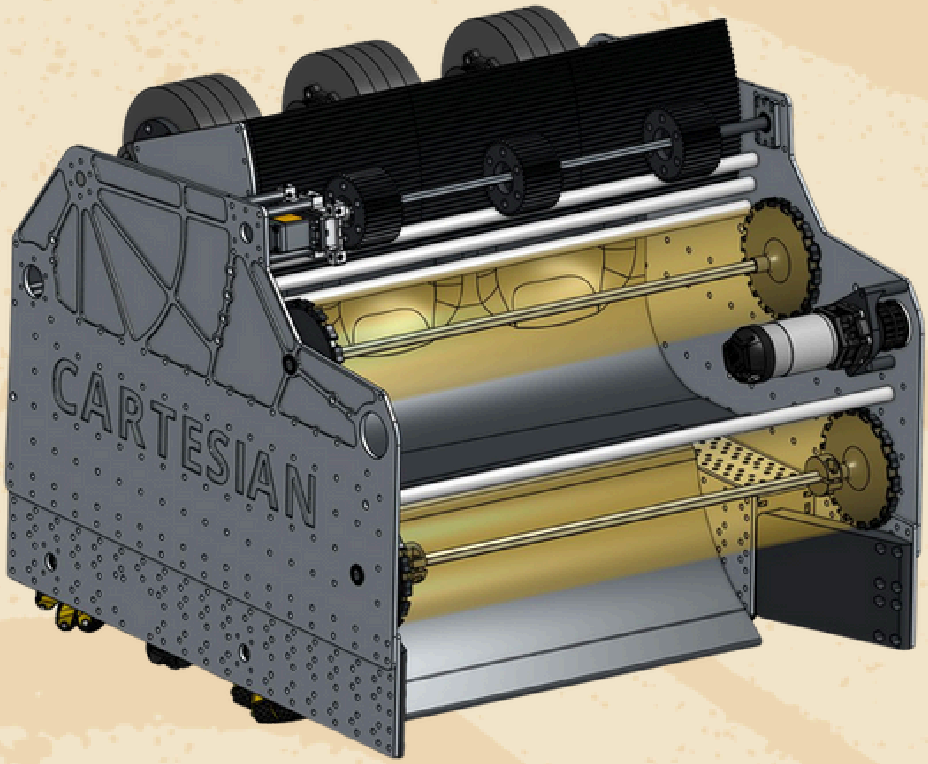
FIRST AGE presented by Qualcomm



CARTESIAN ROBOTICS

#25153

Engineering Portfolio

DECODE presented by RTX

I THINK, THEREFORE I CAN!

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Team Overview

I think, therefore I can.

We are Cartesian Robotics, a dynamic and passionate team from METU Development Foundation Schools, from Ankara, Turkey.

Past Achievements

Since our rookie year in 2023, Cartesian Robotics has rapidly grown from a passionate group of students into an award-winning FTC team. With each season, we've challenged ourselves to think deeper, build smarter, and collaborate stronger. Our journey is marked by constant improvement, international experience, and national recognition.

CENTERSTAGE presented by RTX

- Think Award 3rd Place, Greece National Championship
- Finalist Alliance, İstanbul Off-Season I
- Finalist Alliance, İstanbul Off-Season II
- Design Award 2nd Place, İstanbul Off-Season II
- Innovate Award, İstanbul Off-Season II

INTO THE DEEP presented by RTX

- Winning Alliance Captain, Turkey Championship
- Think Award, Turkey Championship
- Control Award, Üsküdar Regional I
- Inspire Award 2nd Place, Üsküdar Regional II
- Finalist Alliance, Üsküdar Regional II

Season Goals

DECODE presented by RTX

This season, we are committed to growing through continuous learning, embracing challenges as opportunities to improve both as individuals and as a team. With the core value of Gracious Professionalism at the heart of everything we do, we aim to compete with integrity, respect, and kindness.

Our ultimate goal is to succeed once again in regional tournaments, become champions of the Türkiye Championship, and proudly represent our country on the global stage at the World Championship in the United States.

Mission, Vision and Goals

Our **mission** is to empower middle school students through hands-on STEM experiences while fostering creativity, collaboration, and responsibility. By embracing the core values of FIRST—including Gracious Professionalism—we aim to create a learning environment where students can explore freely, compete fiercely, and grow with kindness, respect, and empathy.

At Cartesian Robotics, our **vision** is to empower young minds to become confident problem-solvers and future innovators by combining theoretical knowledge with real-world applications.

Our **goal** is to create a dynamic and inspiring environment where students not only design and build technically advanced robots but also develop essential life skills.

We aim to:

- Build technically advanced and creative robots each season,
- Foster a love of STEM in our community through outreach,
- Grow as a team that supports each other with kindness and respect,
- Encourage leadership, curiosity, and resilience in all members.

We believe that robotics is not just about building machines, but about building character, solving real-world problems, and growing together as a team. Through our efforts, we aim to spark curiosity and inspire the next generation of changemakers. At the heart of everything we do lies our motto: "**I think, therefore I can.**"

Team Structure

Cartesian Robotics is a well-organized team guided by lead & young mentors, teachers, and supportive families, with key departments working together to drive the success of each season. Our team structure operates through a mentorship-driven and collaboration-focused model. Each department is guided by one or more young mentors from our FRC team, who provide technical expertise and leadership throughout the season under the guidence of lead mentors.

Software Team

Autonomous Coding
Sensor Integration
Driver Control Systems

Mechanical Team

CAD & 3D Printing
CNC Manufacturing
Structural Design

Scouting Team

Game Analysis
Strategy Development
Alliance Decision-Making

PR Team

Social Media Management
Sponsorship & Outreach
Event Organization

Driver Team Robot Operation & Match Coordination

Subteams regularly meet both in our workshop and online, holding weekly process meetings to track progress, troubleshoot challenges, and plan next steps. Every team independently documents their workflow, decisions, and development journey—ensuring that each contribution is recorded and reflected in our collective engineering process. This structure allows us to stay organized, promote ownership, and foster continuous learning across all levels of our team.

Sustainability

- At Cartesian Robotics, sustainability means **preparing for the future**—both financially and organizationally—by ensuring our team can thrive beyond a single season.
- Financially, we seek **stable and responsible growth** through sponsorships, partnerships, and community support, ensuring long-term independence.
- We focus on building a legacy of continuity, where knowledge, leadership, and passion are passed from one generation of students to the next.
- Our commitment to sustainability reflects the core values of FIRST: practicing Gracious Professionalism, inspiring future innovators, and creating a community where learning continues for years to come.

Financial Sustainability

At Cartesian Robotics, financial sustainability is secured through the guaranteed support of **our school, the dedication of our families, and the contributions of our sponsors**, all reinforced by careful planning and tracking.

Structured Financial Planning

- At the beginning of each season, our PR Team and mentors prepare a detailed budget plan.
- Each department submits its needs list, which is compiled into a unified document.
- A sponsorship package and the finalized needs list are shared with potential sponsors and families.

Team Budget	Item	Amount
Registration (Annual fee for FTC competitions)	-	5000 $ + 60 000 ₺
Game Set (Field parts and game materials)	Decode Game Set	485 $
Supplies (Main robot controller, sensors, & build kit)	Kit Parts / Control System	800 $
Supplies (Extra / replacement parts)	Motors / Servos	400 $
Supplies (Metal, frame parts, wheels, wiring, etc.)	Hardware / Build Materials	700 $
Events & Outreach (Posters, banners, sponsor materials)	PR Materials	250 $
Events & Outreach (Display boards, signage for competitions)	Pit Display / Branding	100 $
Safety (Gloves, goggles, first aid kit, etc.)	Safety Equipment	60 $

Guaranteed Foundations

School Sponsorship

Our school is our primary sponsor, covering all official FTC registration and competition fees.

Families' Support

Travel, accommodation, and food expenses during competitions are fully covered by our members' parents.

Technical Needs

Robot construction materials, tools, and spare parts are covered by our sponsors, either by directly providing the equipment or by supporting us through financial contributions.

Partnerships & sponsors

At Cartesian Robotics, our success is driven by a strong network of sponsors, partners, and families who sustain our journey through their unwavering support, resources, and commitment to youth-driven innovation.

Ergün

YAMANTÜRK VAKFI

ÖZTEKİN Beta WELL DONE

FİKRET YÜKSEL EĞİTİM VAKFI FİKRET YÜKSEL FOUNDATION

KEMRON Kimya San. Ve Tic A.S.

KOZKA İnşaat

Sustainability

From Recruitment to Worlds: Our FTC Journey

Our season unfolds through a carefully planned timeline where recruitment, training, outreach, and competitions build on each other, guiding us from local beginnings all the way to the global stage.

Recruitment — June

New members begin applications, and evaluation tasks are assigned to candidates.

Preparation — July August

Candidates complete their assigned tasks, participate in training sessions, and finalize their department selections.

September — **Official Kick-off**

With the official Kick-off, team began analyzing the new game, brainstorming strategies, and rapidly prototyping our first ideas.

October — **Towards the Tournament**

The team enters the build phase, conducts mock matches, refines strategies, mentors rookie teams, carries out planned outreach projects, and prepares the portfolio and presentations for judging.

Regionals — November

The team competes in regional tournaments.

Nationals — December

Türkiye Championship with a new iteration of our robot with the lessons learned from regionals.

January April — **Progress**

Robot upgrades, scrimmages, and award preparations, while continuing existing outreach projects and launching new initiatives.

World Champ. — May

Competing on the global stage while sharing our innovations and culture with teams from around the world.

Premiere Event — July

Premiere event participation.

Team Continuity

At Cartesian Robotics, continuity is more than keeping the team alive—it is about cultivating a system where **knowledge, skills, and leadership flow naturally across generations**.
By combining recruitment, mentorship, and leadership, we have built a frame of continuity. The following keystones illustrate how these principles are implemented in our journey.

Large Talent Pool: As a big school community, we receive over 60 applications each year. To sustain quality and fairness, we implement a selective recruitment process that evaluates candidates through assigned tasks and interviews.

Pathway to FRC: We view our FTC team as the foundation for our school's FRC program. FTC not only introduces students to engineering and teamwork but also prepares them to become capable and confident members of our FRC team.

Young Mentorship Model: Many of our graduates transition into the FRC team, and in turn, they return as young mentors to guide FTC students through technical challenges, outreach activities, and competition preparations. This cycle ensures that experience is never lost.

Through this layered system, Cartesian Robotics secures not only its own sustainability but also strengthens the bridge between FTC and FRC—guaranteeing that today's rookies become tomorrow's leaders.

Risk and Constraint Management

Funding Risk	Our participation fees are guaranteed by our school, while travel and accommodation expenses are secured through parental support. For robot parts and materials, we diversify our sources by receiving both direct financial sponsorships and in-kind donations from partners.
Graduation Risk	Since team members graduate after middle school, knowledge transfer is essential. We maintain a structured training ladder where experienced members guide juniors step by step, preparing them to take over critical roles. This layered mentorship guarantees continuity.
Technical Risk	To prevent the loss of technical knowledge, our team documents all code, CAD designs, and mechanical iterations in detail. These records are stored in shared platforms, allowing new members to quickly learn from past designs and continue development seamlessly.

Outreach & Community Impact

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At Cartesian Robotics, outreach is not just an activity — it is the heart of our mission. We believe that every student deserves the chance to experience the **excitement of STEM** and the **values of FIRST**. Guided by the principles of Gracious Professionalism and Coopertition, we design programs that combine kindness, respect, and collaboration with ambition and innovation. Through these efforts, we aim to recruit, inspire, and empower new members of the community to join the global FIRST family and carry its values into the future.

Outreach Objectives

Recruit and support new teams at every level of FIRST.

Introduce robotics to students who have never experienced them before.

Engage mentors and coaches who bring fresh energy into the community.

Inspire young learners to see themselves as problem-solvers, innovators, and changemakers.

We are not only expanding participation in FTC but also building a pathway into FRC, ensuring continuity of STEM opportunities across Turkey. Our contributions—whether in resources, mentorship, or training—reflect our commitment to spreading the **values of FIRST, Gracious Professionalism, and Coopertition** far beyond our own school walls.

Planting Roots, Spreading FIRST

At Cartesian Robotics, our pilot team initiative was born from a simple vision: **to recruit new teams, introduce STEM to fresh audiences, engage new mentors, and inspire future changemakers.** Each year, we establish and mentor an FTC team in a different region, equipping both students and coaches to build a sustainable robotics culture.

Our pilot team founded under Cartesian Robotics, **25601 Pioneers** was created to introduce new communities to FIRST values and competitions.

Mersin – Rebuilt

Our journey began at METU Development Foundation (DF) Mersin Schools, where we supported the creation of an FTC team that competed at the Üsküdar Regional in last season. Graduating students from this initiative are now founding a rookie FRC team in Mersin, preparing to represent their city in this season.

İzmir – Decode

This year, the Pioneers program expanded to METU DF İzmir Schools. Cartesian Robotics provided essential resources, including a full FTC field, electronics, chassis, and motors. We also hosted the mentors in our workshop, training them in areas such as:

- FTC manuals, awards, portfolio, and outreach
- Coding fundamentals, required programs, and using platforms
- Robot construction and competitive design principles
- Team management, sponsorships, and establishing robotics culture

With these foundations, 25601 Pioneers (Current name: OdtümusPrime) are set to compete at Piri Reis Regionals 3 and 4 this season.

İstanbul, Kemerburgaz – Decode

In parallel, Cartesian Robotics took the lead in founding 32012 Okeanos Robotics. We welcomed their mentors into our workshop, provided technical and organizational training, and later visited their workspace to guide the students directly. Thanks to this support, Okeanos Robotics will proudly compete in Piri Reis Regionals 2 and 4.

Looking ahead, we plan to expand this model to more cities across Turkey, with the specific goal of launching a new FTC team in Denizli next season.

Outreach & Community Impact

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STEM on Air: Radio ODTÜ Collaboration

"From the field to the frequency — sharing FIRST with everyone."

As part of our mission to make STEM and FIRST more visible, we launched a three-episode radio series with Radio ODTÜ to share our story and inspire new audiences through sound. Only the first episode has been recorded so far, introducing FIRST and our team's journey to listeners across Turkey.

radyo odtü 103.1

Episode 1: What is FIRST?

In the first episode, we introduced the robotics culture of METU Development Foundation Schools and explained how Cartesian Robotics unites multiple FTC and FRC teams under one vision. We discussed the philosophy of FIRST and its core values —Gracious Professionalism and Coopertition—showing how these principles guide our students to design, build, and grow as future innovators.

Upcoming Episodes

Episode 2 will explore team structure and women in STEM.

Episode 3 will highlight our awards, Houston experiences, and future goals.

The program aims to reach new students, teachers, and parents—spreading the voice of FIRST beyond classrooms and into everyday life.

Empowering Young Minds

Through Access, Technology, and Inspiration...

In line with our outreach mission to make STEM accessible to everyone, Cartesian Robotics led a large-scale computer donation initiative that helped schools across Ankara establish new technology and robotics classrooms.

When our school renewed its computer labs, our PR Team took action to repurpose the old computers for students in need. During the selection process, we **carefully considered different educational levels, accessibility, and socio-economic conditions** to ensure the project reached schools where technology could make the greatest difference.

After contacting multiple institutions, we distributed:

- 10 PCs to Şehit Emrah Yiğit Kindergarten
- 2 PCs to Yenikent Primary School
- 25 PCs, 3 laptops, a projector, and a printer to Çubuk Fatih Sultan Mehmet Vocational & Technical High School (home of Gamma Robotics, Team 10471)
- 20 PCs to Özel Gönüller Public Education Center
- 10 PCs and a projector to METU Yuva Preschool

Through these efforts, five schools gained access to functional technology labs where hundreds of students can now experience STEM learning firsthand.

Early Robotics Education

Taking our collaboration with METU Yuva Preschool one step further, we launched an Early Robotics Education Program to introduce 6-year-olds to the basics of algorithms and robotics thinking.

We equipped the classroom with 3 LEGO Boost sets and began biweekly sessions where our team mentors guide children in building and programming simple robots — **planting the seeds of curiosity and creativity at an early age**.

Many students from METU Yuva later continue their education at our school, meaning this project not only introduces children to robotics early but also **creates a natural bridge for future team members and mentors, strengthening the long-term sustainability of Cartesian Robotics' STEM ecosystem.**

Strengthening the FTC Community

As part of our mission to support and grow the FTC ecosystem in Turkey, we regularly collaborate with other teams by providing mentorship, technical guidance, and resources.

We hosted Team Chaotics (#25564) in our workshop after they lost their mentor, providing robot parts and hands-on technical guidance as they prepared for the European Premier competition in the Netherlands.

We welcomed Robest Robotics (#25156) to our workshop for a collaborative session, offering mentorship and donating a 3D printer and robotic parts to support their upcoming season.

We met with SANA Robotics (#24697), an international team from Kazakhstan, to share outreach experiences, exchange ideas, and discuss future cross-border collaborations focused on STEM accessibility and community engagement.

We organized a joint meeting with AG Robotics (#24230) and JR. AG Robotics (#28743) to evaluate the current FTC season and exchange ideas.

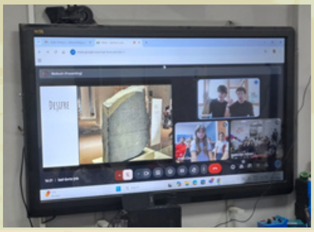
Beyond Robotics: Building Bridges

At Cartesian Robotics, our team plan is built on the belief that sustainable impact comes from learning, connecting, and giving back to the STEM community. In line with our mission to inspire innovation, foster curiosity, and empower young minds, we have established clear goals that guide our actions throughout the season:

Fostering Growth through Expert Collaboration

We actively connect with STEM and engineering professionals and the FIRST community who share their expertise with our team. Through these mentorship sessions, our members gain real-world insights, learn advanced problem-solving techniques, and strengthen their engineering mindset. To enrich this experience, we took part in various expert-led workshops and meetings that contributed directly to our growth.

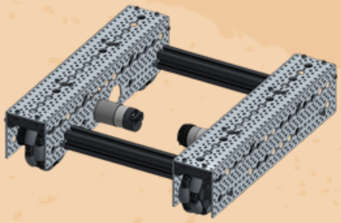
- **Assoc. Prof. Dr. M. Mert Ankaralı** shared the design and development process of the lunar vehicle planned for the Moon's South Pole, giving us a real-world perspective on space engineering.
- **Assoc. Prof. Dr. R. Gökberk Cinbiş** introduced our members to programming and computer vision, showing how AI enhances robotic perception.
- **Yiğit Efe Erdoğmuş**, an FRC Türkiye and World Dean's List Award recipient, inspired us by sharing his FIRST journey and personal insights.
- **Mehmet Ali Uçar**, founder of NFR Products, guided us in robot mechanics, institutionalization, and organizational development.
- **Bedirhan Bal and Timur Sipahi**, archaeology students, helped us explore decoding methods and technologies in archaeology, linking them to this year's Decode theme.
- **Mert Alptoğan**, a former team member and now a civil engineering student at TED University, conducted a CNC workshop for our members, teaching precision manufacturing techniques and their applications in robotic design.

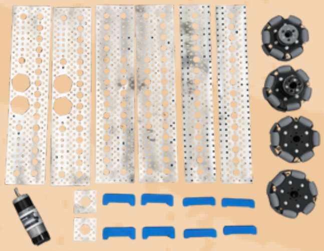


STEM Kits for Future Mentors

In line with our vision of building bridges through science, we launched an educational kit project to expand robotics education beyond our own community. Aimed at teacher candidates in education faculties, the project seeks to inspire future mentors who will bring robotics to new environments and spread STEM learning nationwide.

We designed a five-module robotics education kit, covering key systems such as drivetrain, gearbox, telescopic arm, gripper, and intake. Each module teaches essential mechanical and conceptual principles, combining hands-on learning with competition-level engineering.

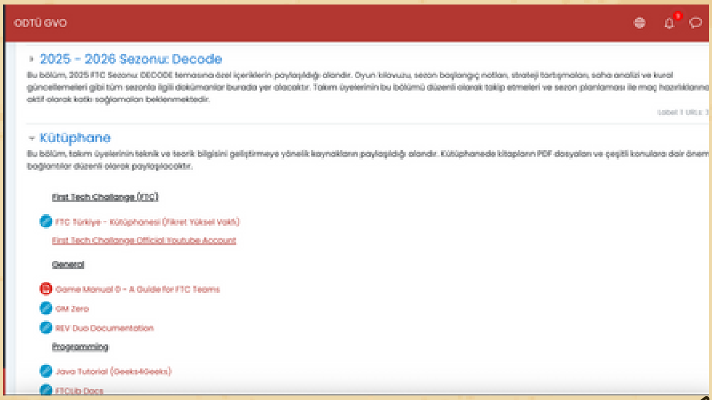




Thanks to our sponsor, Sonuç Yayınları, we built two full kits, which will be delivered to METU Computer Education and Instructional Technology (CEIT) students. Alongside the kits, we created a training curriculum to help educators use them effectively. Fully assembled, the robot is not just a teaching model—it's a functional, FTC/FRC-standard system that helps teacher candidates develop future-ready skills and promote robotics education across schools that will work in Turkey.

Supporting Growth through Our LMS Platform

We created an LMS-based learning platform to support our members' continuous growth and align with the Decode theme by promoting self-directed exploration. This system enables students to progress at their own pace through interactive courses on FIRST values, programming, mechanics & CAD, and portfolio design, helping them decode complex concepts while building both technical and creative confidence.

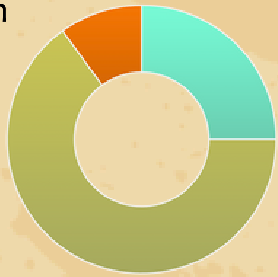


Decoding the Game: Patterns, Points & Possibilities

In the world of FTC, every season begins with a puzzle waiting to be solved. For us, Decode wasn't just a theme — it was a mindset. Before building a robot, we focused on deciphering the game's logic, identifying scoring patterns, and uncovering where each second and point truly mattered. By analyzing **Autonomous, Tele-Op**, and **Endgame** phases, we aimed to transform complexity into clarity — translating data into strategy. Understanding the nuances of **Ranking Points, Pattern bonuses**, and **Cycle efficiency** allowed us to decode not only the field but also the winning formula hidden within it.

1.1 Game Analysis & Strategic Decisions

At the start of the season, we conducted a data-driven game analysis to shape our strategy by examining the scoring contributions of the Autonomous, TeleOp, and Endgame phases. By considering the manual's scoring distribution and calculating the average cycle time achievable with our chassis, we estimated the percentage-based point contribution throughout the match.



%25 Autonomous
Possible differentiation, but limited potential

%65 TeleOp
The main phase where scoring advantage is created

%10 Endgame
Low risk-to-reward ratio

From these numbers, we decoded that:
"Consistency beats risk."

The TeleOp phase proved to be the defining part of the game. Although Pattern and Ranking Points (RP) appear advantageous, they consume significant resources, and the required modifications to core subsystems could potentially increase cycle time. Therefore, we decided to attempt Pattern completion only if it could be integrated without slowing subsystems or consuming additional resources.
Our goal was not to reach the maximum theoretical score, but to approach it in the most efficient way possible.
The same reasoning guided our Endgame decisions: achieving near-maximum results without developing a new subsystem was the smarter and more sustainable engineering approach.

1.2 Ranking Points Philosophy

The new Ranking Point (RP) system introduced in this year's FTC Decode season directly influenced our strategic mindset.Our mentors, with their FRC background, reminded us of a timeless principle:

"Earning RPs matters, but the best RP comes from winning the game itself."

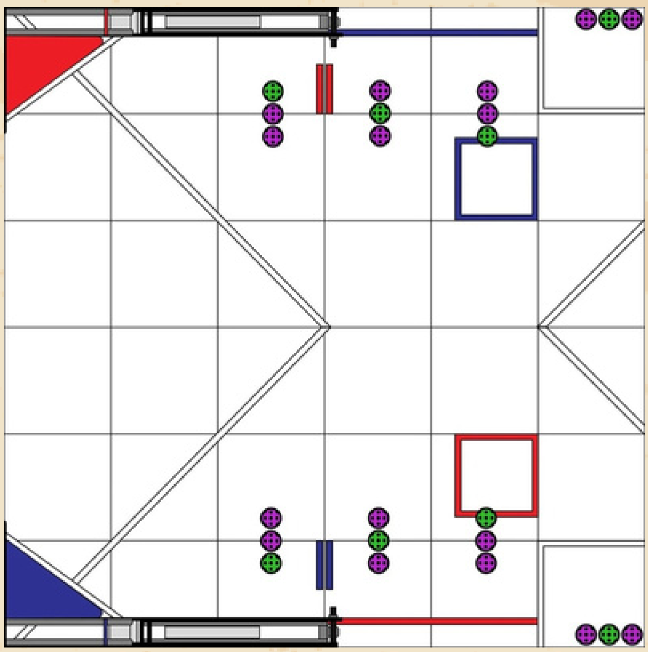
This philosophy guided us to never alter our core mechanisms solely for RP gains. Instead, **we optimized our robot** not for a single task — but **for the entire flow of the game**, ensuring that every subsystem contributed to consistent, all-round performance.

1.3 Cycle of the Decode

To measure efficiency, we divided the game into **robot cycles**—each representing a full loop from collection to scoring.
By analyzing driving paths, distances, and mechanism speeds, we identified three core cycle types:

- **Short Cycle (≈4–5s):** Rapid 3-ring pickup and shot from nearby zones.
- **Mid Cycle (≈6–7s):** Quick reposition with one obstacle or direction change.
- **Long Cycle (≈9–10s):** Includes Pattern shots or field re-alignment.

Using these data points, we modeled **average realistic cycles per match** and compared total scoring outputs across different autonomous and teleop strategies.
Rather than chasing the maximum theoretical score, we aimed for the highest efficiency threshold.



Blueprints of Thought: Engineering our Mindset

Every great robot begins with a philosophy. Before opening CAD or cutting metal, we asked ourselves a simple question:

what should our robot stand for?

The answer became the foundation of our engineering journey: a set of design values shaped by years of FRC experience, NASA's engineering mindset, and Cartesian Robotics' belief in logical creativity. These principles guided not just what we built, but how and why we built it.

Design Values & Origins

During the summer, our FTC and FRC teams came together with mentors who have competed in FRC since 2018 to create our Design Manifesto. Inspired by **NASA's engineering principles**, we defined the core philosophies that became the compass of every design decision we made this season. Each subsystem — from our drivetrain to our intake and shooter — was engineered under these guiding values, ensuring that every iteration reflected logic, not luck.

Core Design Phiosophies

Icon	Philosophy	Explanation	Effect on Design
	K.I.S.S. (Keep It Simple & Straightforward)	Simplicity accelerates manufacturing and maintenance.	Avoided unnecessary complexity across all mechanisms.
	Low Center of Gravity	A lower center of mass increases stability.	Heavy components were positioned near the base for balance.
	Touch It, Own It	Whatever you touch, you must control.	Guided our intake design for consistent game-piece control.
	Rollers Rule	Flexible contact often performs better than rigid grip.	Soft roller designs increased intake reliability and tolerance.
	Drivetrain is King	Mobility is the foundation of scoring.	Chassis became the strategic core of our robot.
	Autonomous Domination	Autonomous decisions are faster and more stable than human input.	Simplified driver operations and ensured autonomous consistency.
	Design for Adjustability	Easily adaptable, endlessly tunable.	Every mechanism was patterned for modular replacement and iteration.

"These design pillars ensured that every iteration reflected logic, not luck — Cerberon is the embodiment of these values."

Subsystems / Shooter: The Triple Fire Decoder

Cerberon's shooter system emerged as the result of an extensive cycle of research, testing, and refinement. Although our FRC experience gave team members a strong understanding of shooting physics, the motor limitations and projectile characteristics of FTC demanded a completely new approach.Therefore, we structured our shooter development into three key stages:

Physical Analysis → Experimental Iteration → Adaptation Optimization.

Physical Analysis - Angular Momentum

In the first stage, we identified **angular momentum ($L = I \times \omega$)** as the key parameter governing the projectile's flight stability. The pressure difference created around the spinning ball generates a **Magnus effect**, which stabilizes the trajectory mid-air. To utilize this principle, we designed a secondary hood wheel that imparts additional spin to the projectile. This wheel increases the angular velocity (ω), and together with the ball's moment of inertia (I), amplifies the overall angular momentum (L) — improving balance and consistency rather than raw speed.

The primary goal of this analysis was stability over power. Based on these physical insights, we moved on to the next stages: prototyping and iterative testing to translate the physics into consistent real-world performance.

Experimental Iteration – Compression & Energy Transfer

Throughout our testing phase, we evaluated shooting accuracy through a data-driven mapping process. For every 20 cm of distance, we tested a wide range of hood angle and wheel RPM combinations, recording the outcomes in a detailed shot map (Excel-based table).

By the end of this stage, the shot map was integrated into Cerberon's control algorithm, enabling the robot to automatically detect its position and target, and select the optimal shooting parameters in real time.

1. Compression Prototyping & Iterations

To determine the most efficient compression value for the shooter, we conducted a series of systematic tests.

Compression (mm)	Observation	Result
6 mm	Insufficient energy transfer: projectiles fell short.	❌ Unstable due to surface holes.
10 mm	Inconsistent range; some balls over-compressed.	⚠️ Partial stability.
15 mm	Optimal energy transfer with consistent spin and trajectory.	✅ Selected configuration.

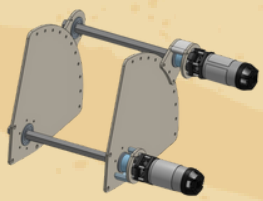
The instability observed at 6 mm and 10 mm compressions was caused by the holes on the ball's surface. When these holes aligned with the shooter wheels, energy transfer became inconsistent, resulting in unstable launches.

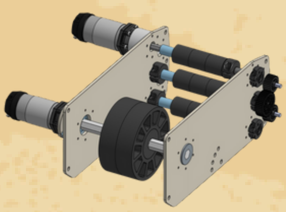
During this process, we analyzed the energy transformation using the formula: $E_p = \frac{1}{2}kx^2$

This showed that as compression distance (x) increases, the stored elastic energy grows exponentially. However, excessive compression led to torque loss within the system. Through iterative testing, we found the equilibrium point at 15 mm, balancing maximum energy transfer with mechanical stability.

2. Motor Gearbox Ratio Prototyping & Iterations

Since our shooter's angular momentum directly depends on rotational speed (RPM), we began with the hypothesis that higher RPM would increase shot stability and range. However, we were also cautious that using a 1:1 gearbox might not provide enough torque. To verify this, we systematically tested 3:1, 2:1, and 1:1 ratios, calculating our target times and velocity values using the re.calc tool throughout the process.





Gear Ratio	Observation	Result
3:1	RPM and resulting momentum were too low.	❌ Insufficient
2:1	Better control, but inadequate momentum at long range.	⚠️ Partial successful
1:1 (6000 rpm)	Contrary to our concerns, torque was sufficient and produced consistent, stable shots.	✅ Optimal

Ultimately, the 1:1 gearbox became our chosen configuration. It delivered the right balance between torque and RPM, allowing Cerberon's shooter to maintain accuracy and repeatability without sacrificing energy efficiency.

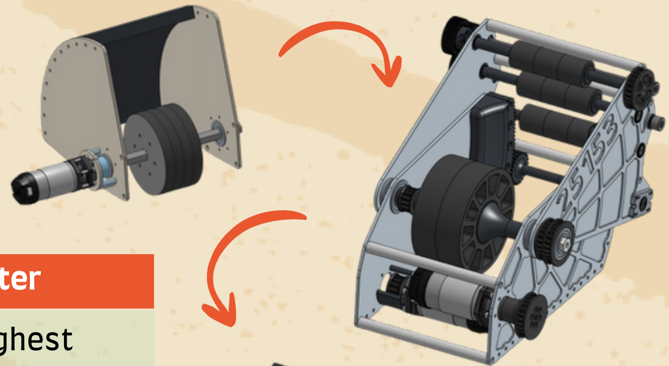
Subsystems / Shooter: The Triple Fire Decoder

2. Motor Gearbox Ratio Prototyping & Iterations

Our initial physical analysis suggested that adding an extra wheel on top of an adjustable hood would allow us to fine-tune the spin applied to the ball, producing the most accurate shot.
Following this hypothesis, we began a series of structured prototypes and tests to validate our assumptions.

Iteration	Configuration	Performance Summary
Fixed-Angle Hood & Single Shooter Wheel	1 DC motor	Achieved ~80% accuracy from a fixed position, but unsuccessful at variable shooting angles.
Adjustable Hood & Shooter Wheel	1 DC motor + 1 × 5-turn servo	Reached ~80% accuracy at close range and ~70% at longer distances.
Wheeled Adjustable Hood & Shooter Wheel	2 DC motors + 1 × 5-turn servo	By tuning both hood RPM and angle, achieved ~95% accuracy from nearly every position on the field.

After extensive testing, the wheeled adjustable hood configuration demonstrated superior adaptability, stability, and precision. By manipulating both hood angle and auxiliary wheel RPM, Cerberon's final shooter achieved 95% shot accuracy across almost the entire field — marking the culmination of our design and iteration process.



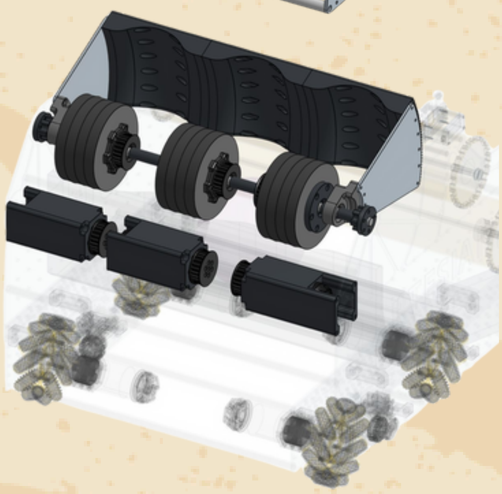
Adaptation Optimization – Transition to the Triple Shooter

Our final wheeled adjustable-hood prototype achieved the highest accuracy, but the motor limitations of our mecanum drive forced us to rethink system complexity. Since high-motor-demand systems like turrets were impractical, and the single-motor flywheel caused long reload times, every improvement came with trade-offs.

Adding a second shooter motor increased firing speed but slowed the indexing mechanism, contradicting our K.I.S.S. (Keep It Simple & Straightforward) philosophy. Although the system worked, it lacked the agility and consistency required for effective pattern scoring.

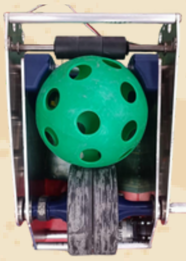
Revisiting our core principles, we reached a clear conclusion:

Given our limited time frame, system complexity had exceeded necessity.



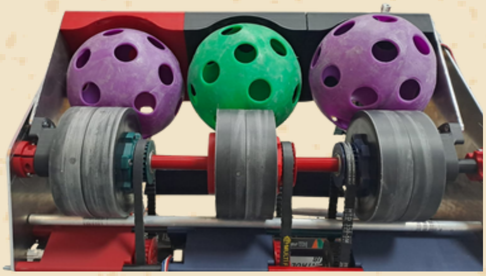
This realization sparked the triple-shooter concept — three independently driven flywheels mounted on a single shaft, each controlled by its own PID loop. This structure enabled simultaneous launches, making reload time negligible and dramatically improving cycle efficiency.

A single adjustable hood maintained accuracy across distances, while the remaining motor powered a full-width intake for faster collection.



A servo-assisted inclined funnel guided balls smoothly into the shooters.

In the end, these refinements gave rise to Cerberon — a robot that is fast, flexible, and perfectly aligned with our Keep It Simple & Straightforward design philosophy.



"We learned that understanding motion physics can be more valuable than building complexity."

Subsystems / Intake: Touch It, Own It

Physical Analysis – Angular Momentum

In the first stage, we analyzed the surface texture and friction coefficient of the game elements. Our goal was to achieve a contact force high enough to grip the objects without slipping, yet low enough to avoid compression or jamming.

Since intaking is the most frequently repeated action during a match, every fraction of a second gained here directly translates to competitive advantage. For this reason, our intake had to be not only fast but also driver-friendly, allowing seamless control and recovery.

To achieve this, we designed a full-width intake system — enabling our robot to make immediate contact with objects anywhere along its front edge and secure them in one smooth motion.
This design embodies one of our core principles:

Touch It, Own It.

Experimental Iteration – Material and Geometry Testing

To make the intake more flexible, we first tested an axially moving design using spring-supported polycarbonate tubes, followed by prototypes with soft wheels and star-shaped rollers. However, the star tips tended to catch in the holes on the balls, reducing consistency and control.

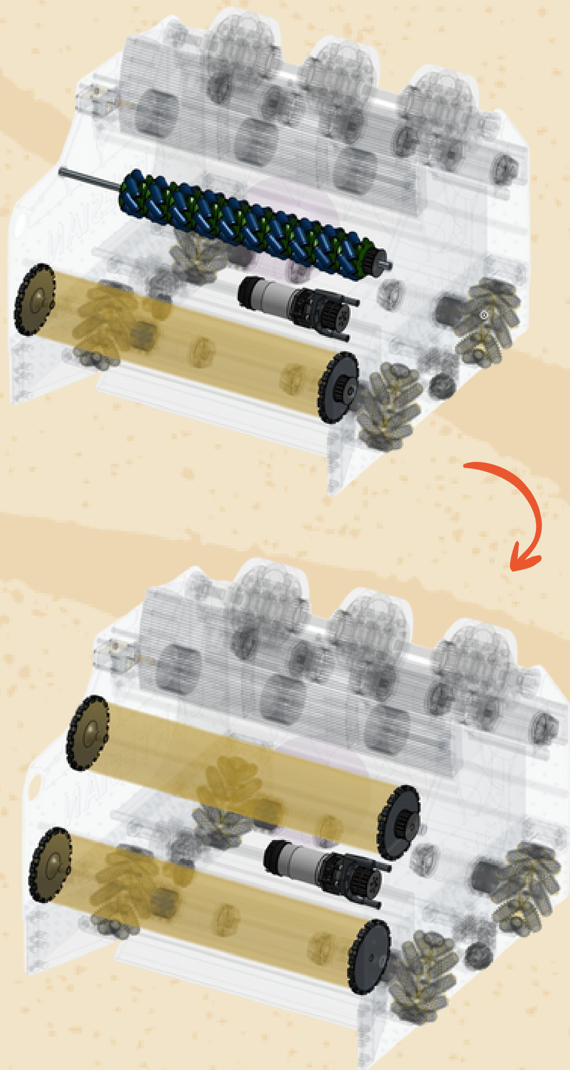
In the final iteration, we developed a rubber-band-based dual-direction tension system. This configuration proved both lightweight and durable, capturing game pieces smoothly and reliably while maintaining high cycle speed.

Adaptation Optimization – Perfection in Simplicity

At the end of all iterations, the intake system achieved a structure that was both lightweight and flexible, with an optimized friction coefficient for secure, consistent handling. The final design delivered reliable collection performance even at high speeds while remaining durable and free from unnecessary complexity.

This solution perfectly embodied our K.I.S.S. (Keep It Simple & Straightforward) philosophy and became one of the cornerstones of Cerberon's overall design identity.

"Simplicity is not about having fewer parts — it's about every part working perfectly."



Subsystems / Funnel Mechanism & Drivetrain

Three-Axis Funnel Mechanism – Ball Alignment System

Physical Analysis - Need for Jam-Free and Balanced Sorting

For the shooter system to operate efficiently, balls needed to align beneath it smoothly, without compression or instability. However, initial hopper designs with flat or single-axis inclined surfaces revealed two major issues:

- On flat surfaces, balls arriving from the same point compressed each other, completely halting flow.
- On single-axis inclines, balls accumulated on one side, and when new ones entered, pressure increased with no escape path.

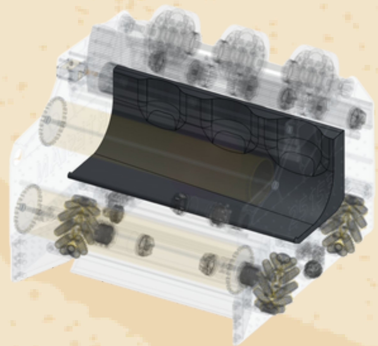
To solve these problems, we needed a design that allowed balls to move forward freely while also providing lateral escape space when necessary — leading to the concept of a three-axis funnel mechanism.

Experimental Iteration – From Mecanum to Passive Geometry

Our first approach used a mecanum roller system powered by the intake motor, aiming to actively guide balls toward the shooter line. However, this setup failed to deliver sufficient pushing force — the rollers couldn't move the balls effectively, while the motor operated under heavy load, overheating and losing performance. This experiment taught us that passive, physics-based designs were more reliable than complex active systems.

In the next iteration, we redesigned the funnel with three-axis inclines, allowing balls to align naturally using their own kinetic energy and gravity, achieving smooth and jam-free positioning.

Adaptation Optimization – Three-Axis Geometric Balance

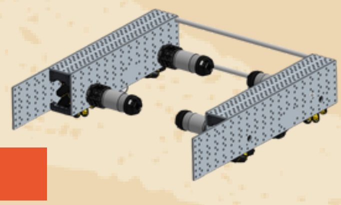


The three-axis incline achieved a perfect balance between flow and stability:

- X and Y axes provided lateral escape paths for balls under pressure.
- The Z-axis optimized the contact surface, balancing friction for smooth motion.

As a result, three balls could align quickly and stably beneath the shooter without compression or jamming. The entire system operates purely through gravity and geometric flow, requiring no additional motors, servos, or sensors. This solution stands as a tangible embodiment of our K.I.S.S. (Keep It Simple & Straightforward) and Design for Adjustability principles.

"Simplicity is not the result of power — it is the result of understanding."



Drivetrain & Structure: Keep It Simple & Straightforward

Physical Analysis

Following the "Drivetrain is King" principle, we focused on mobility and control from the start.

Game analysis showed that quick repositioning was key to scoring, while shared parking space made compactness essential.

The result was a maneuverable, space-efficient chassis optimized for speed and precision in tight areas.

Experimental Iteration

The wide-base design caused parking alignment issues, so we embedded the front wheels to shift the rotation center inward. Adding patterned mounting holes made the chassis modular and cut iteration time by 60%, allowing faster testing. This design also improved maneuverability and adaptability across different field setups.

Adaptation Optimization

Our chassis evolved into a lightweight, agile, and reconfigurable platform. Iteration time decreased by 60%, parking success improved, space was freed for the second robot, and overall performance consistently approached maximum scoring efficiency.

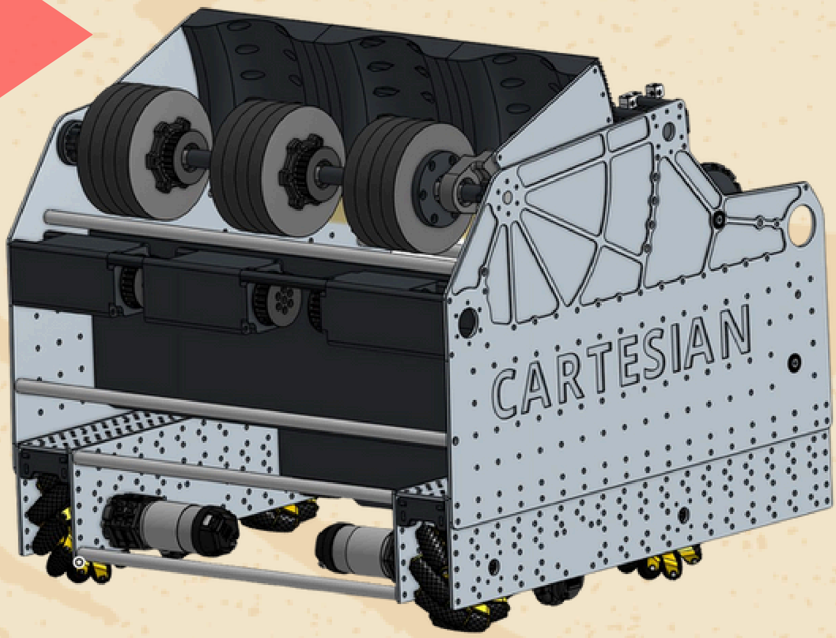
"Adjustability isn't about changing often — it's about being ready when change is needed."

CERBERON - The Triple-Fire Decoder

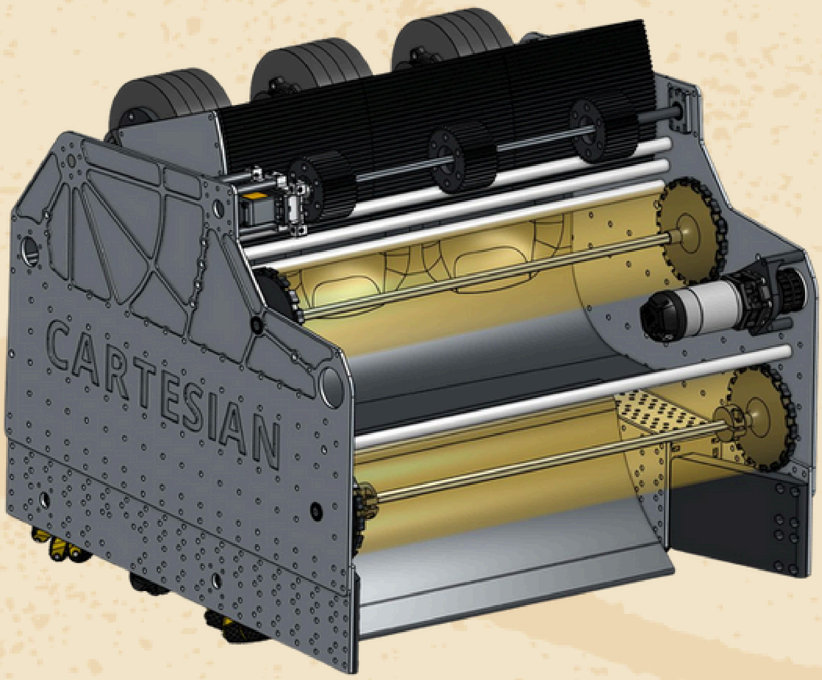
CERBERON

This season, we named our robot Cerebron, inspired by Cerberus, the mythological three-headed guardian, and cannon, symbolizing power and precision. With its three-shooter design, Cerebron mirrors the game's triple-ball element, combining myth and engineering into one fierce identity. Each shooter represents not just a mechanism, but a mindset—accuracy, coordination, and intelligence—working together to decode every challenge on the field. In the spirit of The Age and Decode, Cerebron stands as our triple-headed powerhouse, ready to unleash precision, logic, and innovation in perfect harmony.

3+ Km Filament
50+ Hours 3D Design. **32+ Hours Assembling**
1000+ Line Code



- Custom Cartesian's Chasis: 3mm Aluminium Inner - Polycarbon Outer (Lighter and Smaller)
- Dead Wheel Odometry. Less Precise - Less Surface Dependent)
- 8x REV Hex UltraPlanetary Motor
- 4x Gobilda Servo Motor
- Adjustable Hood with Ball Shaped Path
- Custom Designed Indexer
- Full Size Rubber-band Intake
- 6x 60A 4-inch Flywheel
- Husky Lens and WebCam



- Cerberon is a **three-headed shooter robot** engineered for maximum efficiency — capable of launching three rings simultaneously with precision and speed.
- Its **modular, adjustable design** and gravity-assisted intake system embody our philosophy: **"smart simplicity beats complexity."**
- Cerberon was **born from iteration, failure,** and **decoding**. Abandoning a working shooter for a smarter one taught us more than success ever could.
- We didn't just play Decode — **we decoded it.**

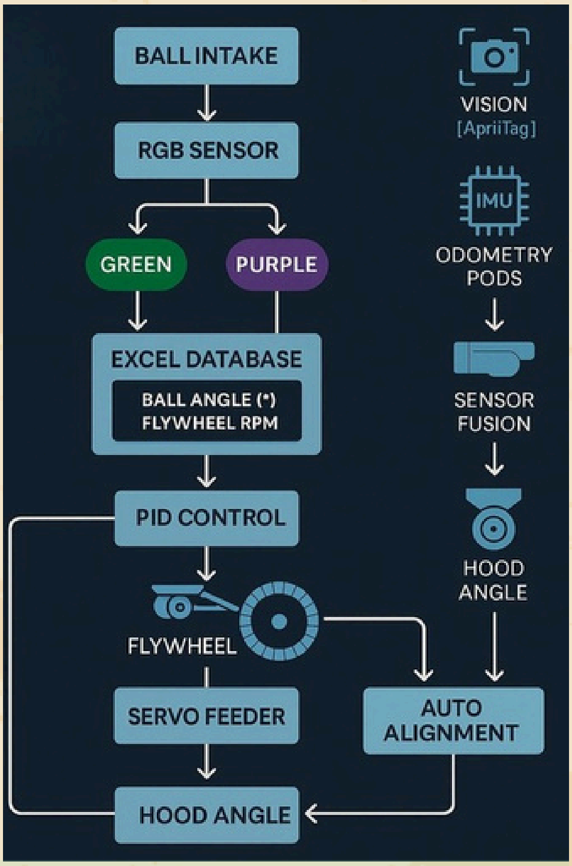
Control: When Logic Takes the Wheel

25153 cartesian

Overview – Harmony of Mind and Mechanics

Cerberon's control system unites mechanical power with intelligent decision-making. Each subsystem — intake, funnel, servo-feeder, shooter, and drivetrain — operates through its own sensors, yet all converge at a central decision hub. This core reads, interprets, and responds to field data in real time, transforming the robot from a command follower into a data-driven thinker.

"Understanding what seems complex is the purest form of engineering." — K.I.S.S.



Color Recognition & Database-Driven Decision System

Cerberon handles two ball colors: purple and green.
An RGB color sensor at the end of the intake line detects each ball's color in milliseconds and sends it to the control software. The robot then selects the appropriate shooter channel and angle/RPM combination from a database built from field-test data. Each database cell stores optimal hood angle (°) and flywheel RPM for a given distance and color. The entire process completes in under 0.5 seconds, giving each ball a personalized shot profile based on position, color, and distance.

"When understood correctly, data is not numbers — it becomes strategy." (Design for Adjustability)

PID Control – Stability in Precision

Each of Cerberon's three shooter motors runs on its own PID loop:
P for rapid response, **I** for long-term balance, and **D** for damping sudden changes. This ensures every shot maintains identical kinetic energy despite friction or ball variation, guaranteeing repeatability and precision.

"Power lies not in hitting the target, but in maintaining balance." (Touch It, Own It)

Sensor Fusion – Vision and Awareness

Localization combines three sensor systems:
Odometry Pods: Track X–Y motion **IMU:** Measures heading. **Camera (Vision):** Reads AprilTags for field reference
By merging these data streams, sensor fusion keeps position error under 2%. Vision automatically adjusts hood angle via PID alignment, ensuring consistent accuracy.

"A robot's sight doesn't come from its camera, but from the harmony of its sensors." (Drivetrain is King)

Servo Feeder & Pattern-Synchronized Control

Each shooter has an independent servo feeder. The color sensor activates the corresponding servo, directing the ball to the correct shooter. Even when all three servos operate simultaneously, priority logic prevents overlap by sequencing tasks based on color data. The same logic runs seamlessly between autonomous and tele-op modes, maintaining data persistence and smooth pattern continuation.

"Harmony doesn't make a robot fast — it keeps it fast." (Design for Adjustability)

The Philosophy of Control

Cerberon's architecture isn't built on perfection, but on balance and continuous learning. Every sensor offers perspective, every loop represents adaptation, and together they create a robot that thinks through decisions rather than just following commands.

"Control is not about commanding a robot — it's about teaching it to think."

ChatGPT was used during the writing and translation process, helping us craft clear, impactful content and seamlessly switch between Turkish and English.

Canva supported us in the design and layout of this portfolio, allowing us to visually organize our content in a creative and accessible way.